Econ 520

Assignment 1

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Question 1:

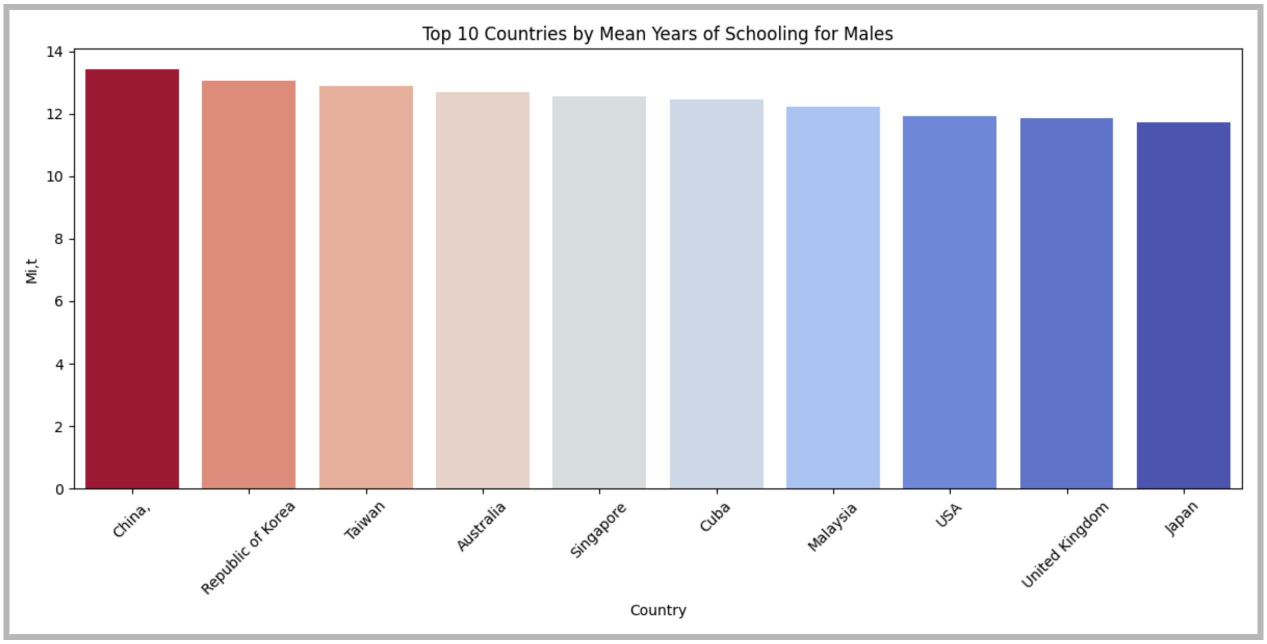
(i) **Patterns in schooling and gender bias in it across countries in 2010:**

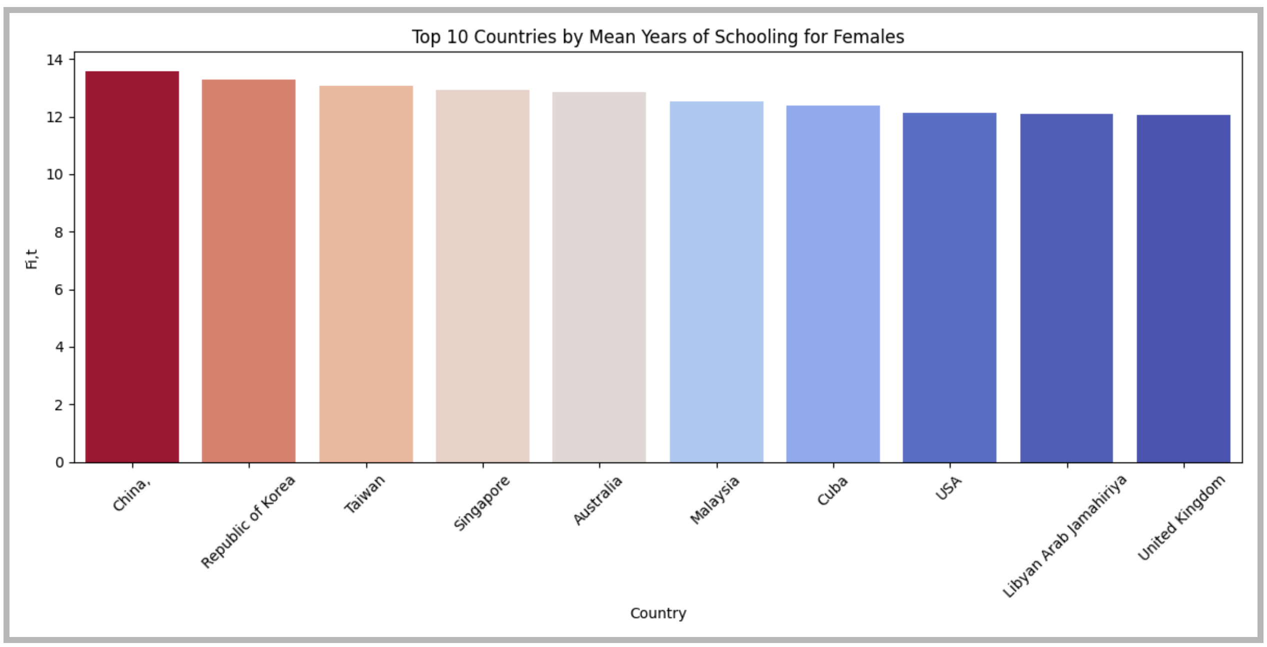
	Fi,t	**Mi,t**	**MFi,t**	**GBSi,t**
Mean	8.907390	8.746048	8.823986	1.005062
Median	9.621000	9.183000	9.455000	0.975067
Std	2.526538	2.193558	2.327604	0.135044
Maximum	13.570000	13.402000	13.487000	1.768583
Minimum	2.085000	2.756000	2.588000	0.671222

The average number of years of schooling for Male(Mi,t), Female(Fi,t) and All people(MFi,t) in the year 2010 is close to 9 years. The average gender bias ratio, which is the ratio of males to females is approximately 1.005. This value indicates that across the country we almost do not have any gender bias in average years of schooling.

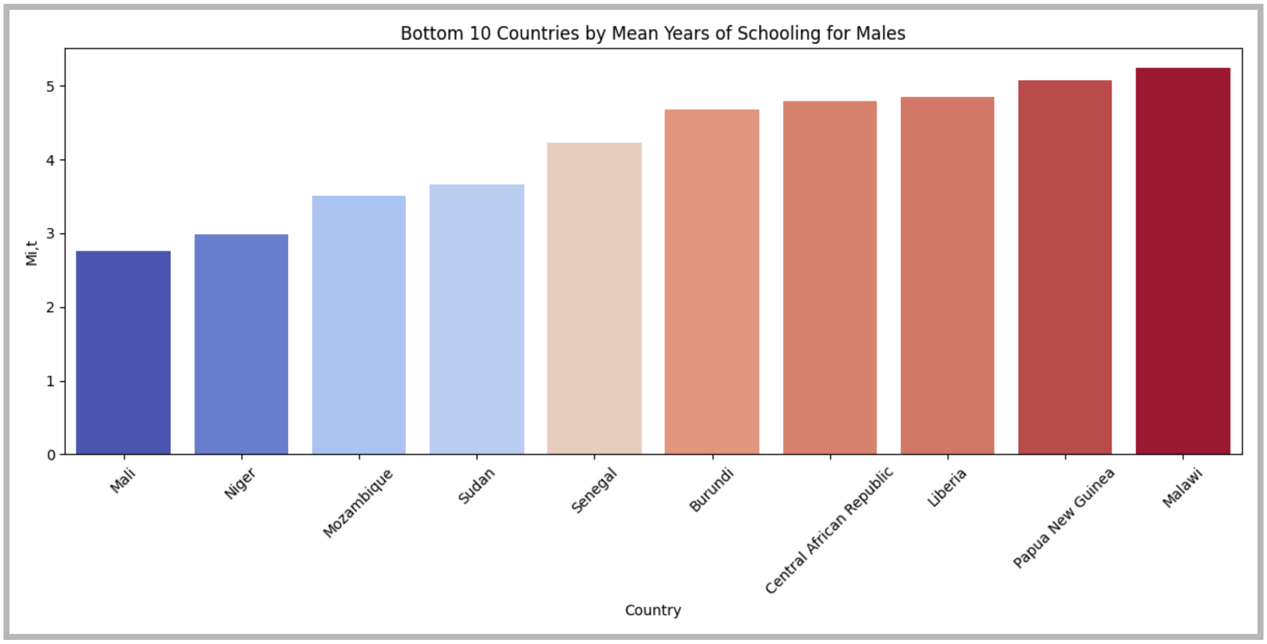
The median values for Male, Female and All persons are 9.18, 9.62, and 9.45 years respectively.The standard deviation for Male, Female and All persons is slightly above 2 which indicates there is moderate fluctuation in schooling years.

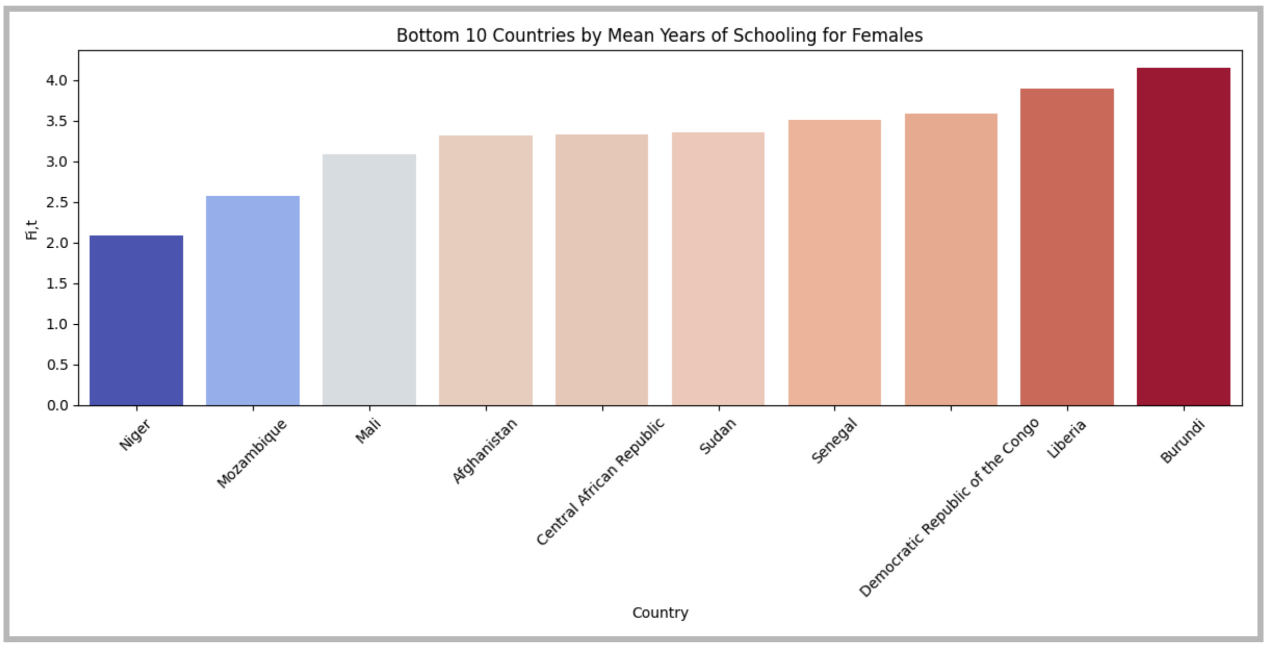
The Maximum values for Schooling is approximately 13.40, 13.57 and 13.48 for Males, Females and All persons and the minimum is close to 2.75, 2.08 and 2.58 years. This number shows the stark difference between the education attainment level between countries. This correlation may also explain the growth rate between countries as seen in our regression analysis.





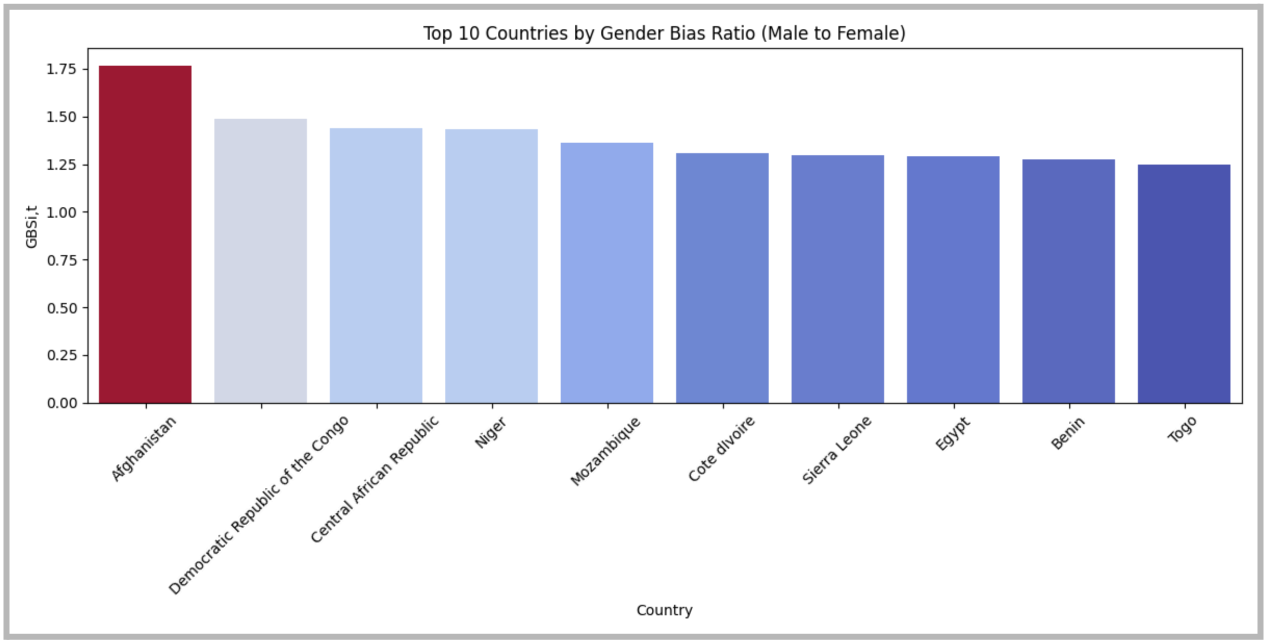
The Bar Chart represents the top 10 countries with the highest mean years of schooling for males and females globally in 2010. The list is dominated by countries from the Asia-Pacific region, which includes China, Korea, Taiwan, Australia, Singapore, and Japan. This indicates the higher emphasis on education in this region. Also USA,Cuba and UK in the list indicate strong educational infrastructure in North America and Europe as well. These countries on average have approximately 12 years or more in average schooling. It shows the emphasis put on education by these countries.

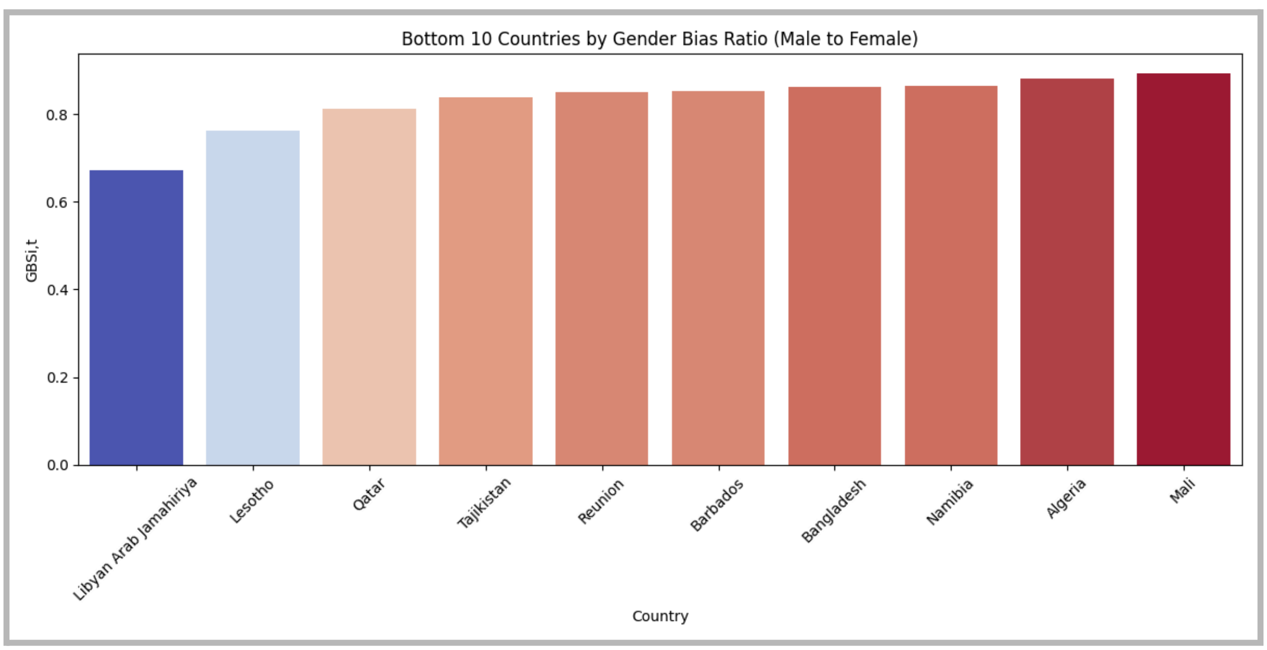




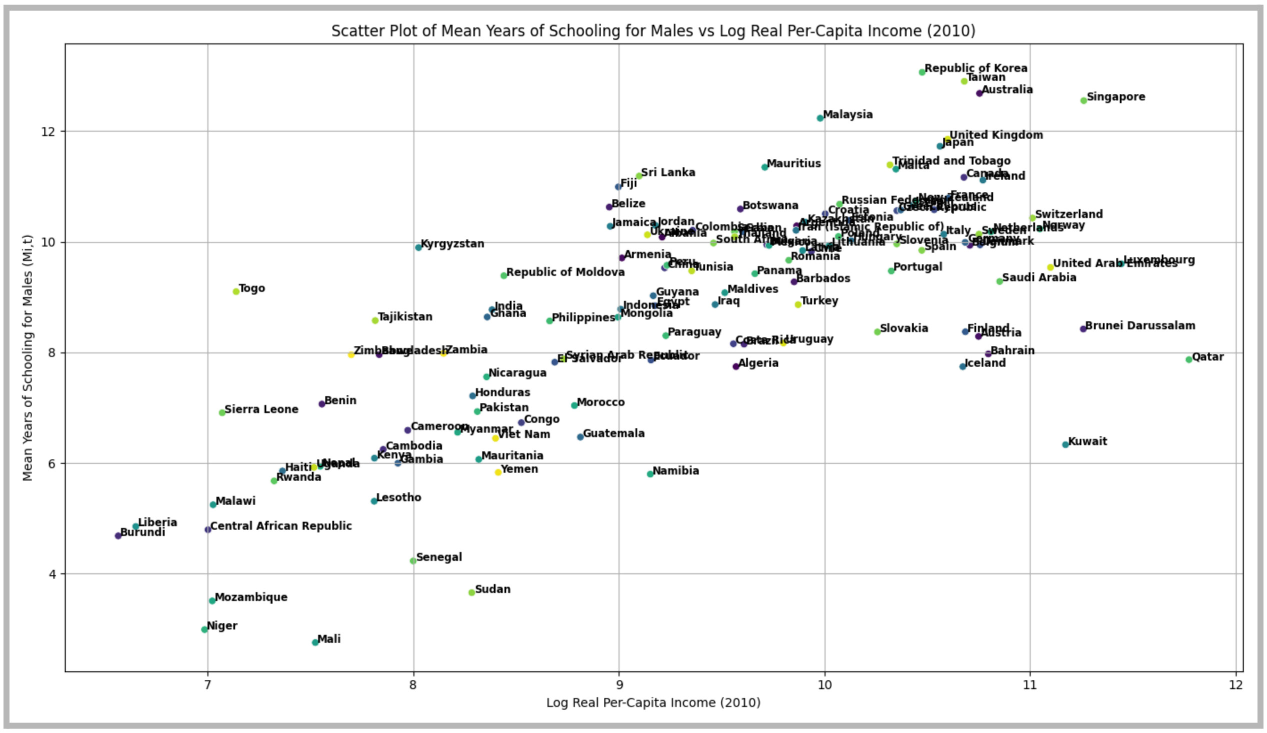
The Bar chart here represents top 10 countries with average lowest years of schooling in 2010 which is in the range of 2 to 5 years.Here the lowest countries have approximately 2 years in schooling among its population. It is really small when compared to the top 10 countries with the highest number of average schooling. The list is dominated by countries from the African region, suggesting poor educational infrastructure and significant challenges for the educational system.

Also conflict and other political factors might contribute to low years of schooling across these countries. The surprising presence of Afghanistan in the lowest years of education for females indicates there might be different socio-political reasons for this.

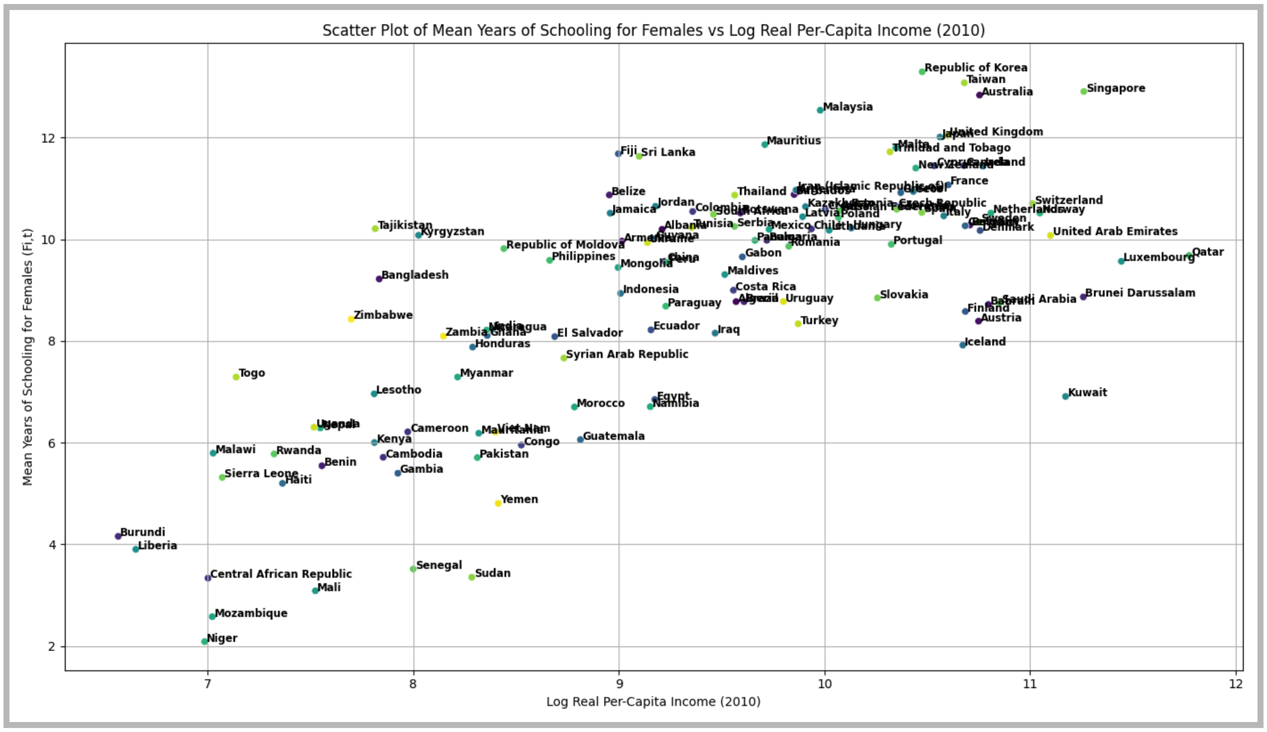




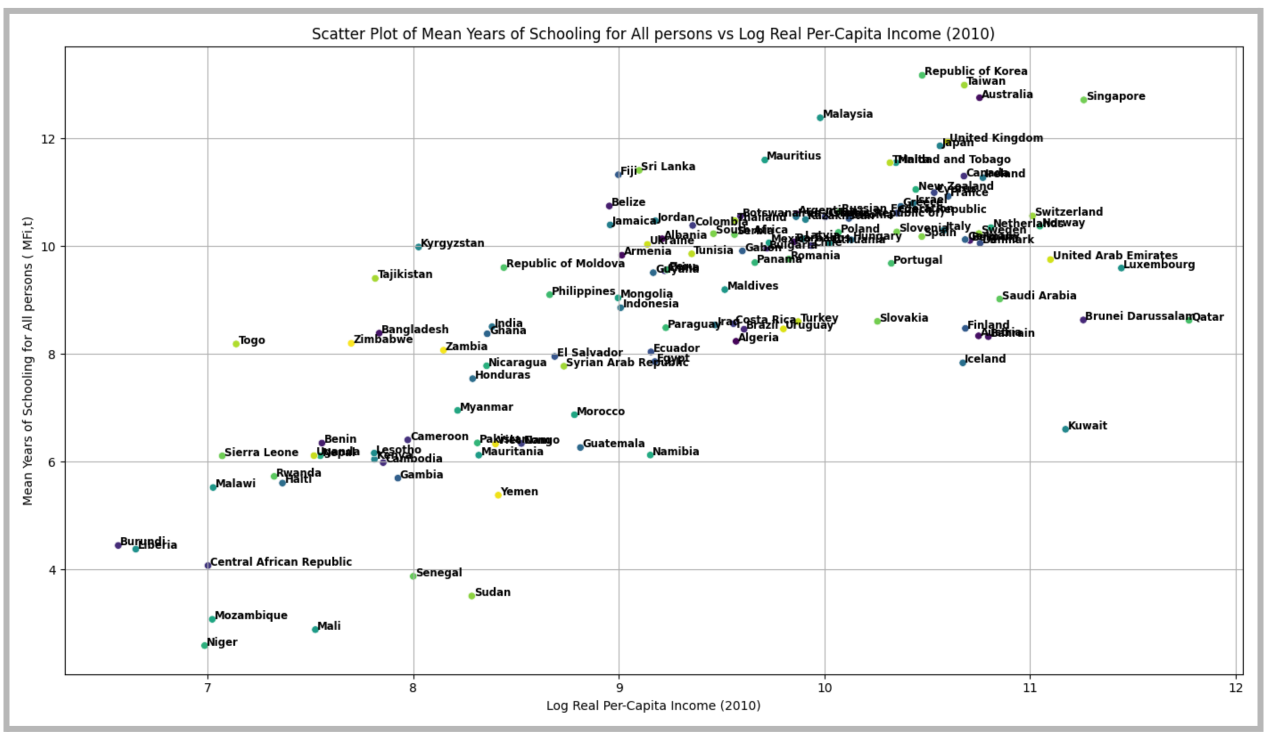
The Bar chart represents the countries with significant gender bias in favor of males in 2010. In the first chart, Afghanistan has the highest bias against females at approximately 1.75 ratio meaning if 1 female goes to school for 1 year then the 1.75 male goes to school for 1 year which clearly indicates a clear preference for male education in these countries. In the second bar chart, Libyan Arab Jamahiriya has the highest bias toward female education.From the chart we can see that African countries show a preference toward male education in general. On the other hand, there is no specific region that has a gender bias towards females as all the countries listed showing female education preference are from different regions.



This scatterplot shows the relationship between average male years of schooling and the real income level of the countries in 2010 which seems to show a positive relationship.



This scatterplot shows the relationship between average female years of schooling and the real income level of the countries in 2010 which also seems to show a positive relationship.



All three scatterplot above shows that there is positive correlation between per capita income and mean years of schooling in 2010. This means that countries with higher per capita income can invest and spend more in educational infrastructure and attainment.We can say that higher economic development is associated with better educational outcomes. However, it can be also argued that higher education can lead to higher real income. Countries from the middle-east such as UAE, Kuwait and Brunei have high per capita income and high mean years of schooling. On the other hand, countries from the African region show lower per capita income and low mean years of schooling. The scatterplots show the importance of investment in education as it may lead to higher per capita income.

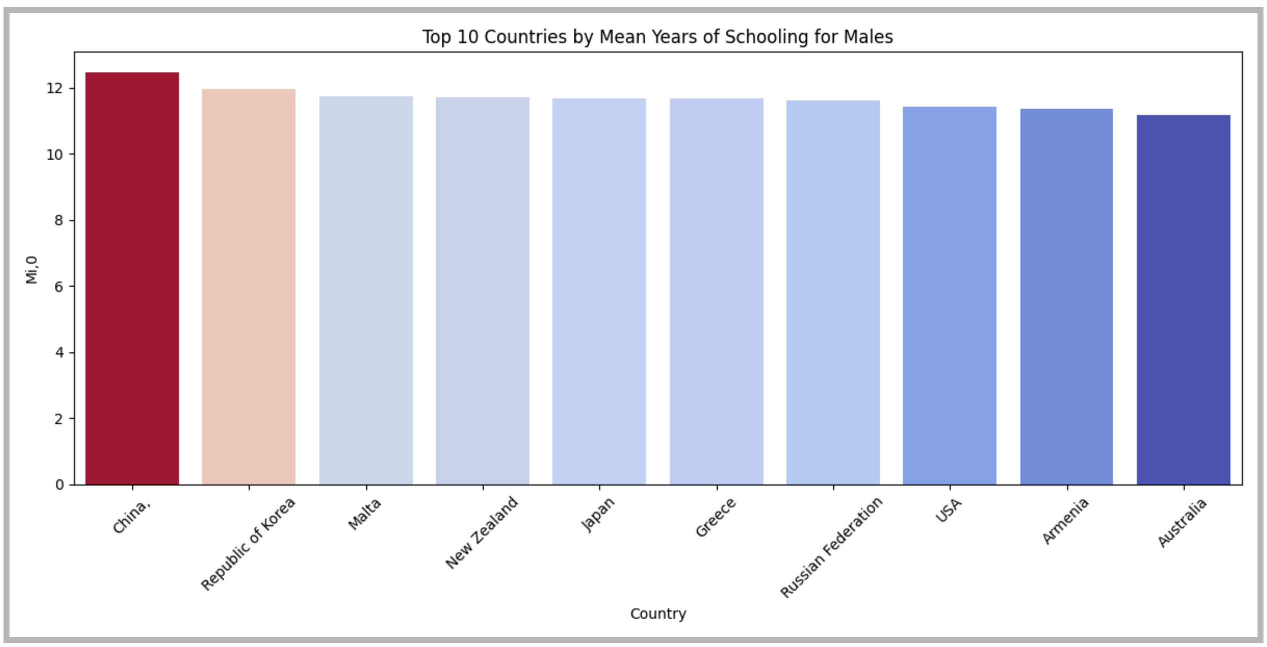
(ii) **Patterns in schooling and gender bias in it across countries in 1990**:

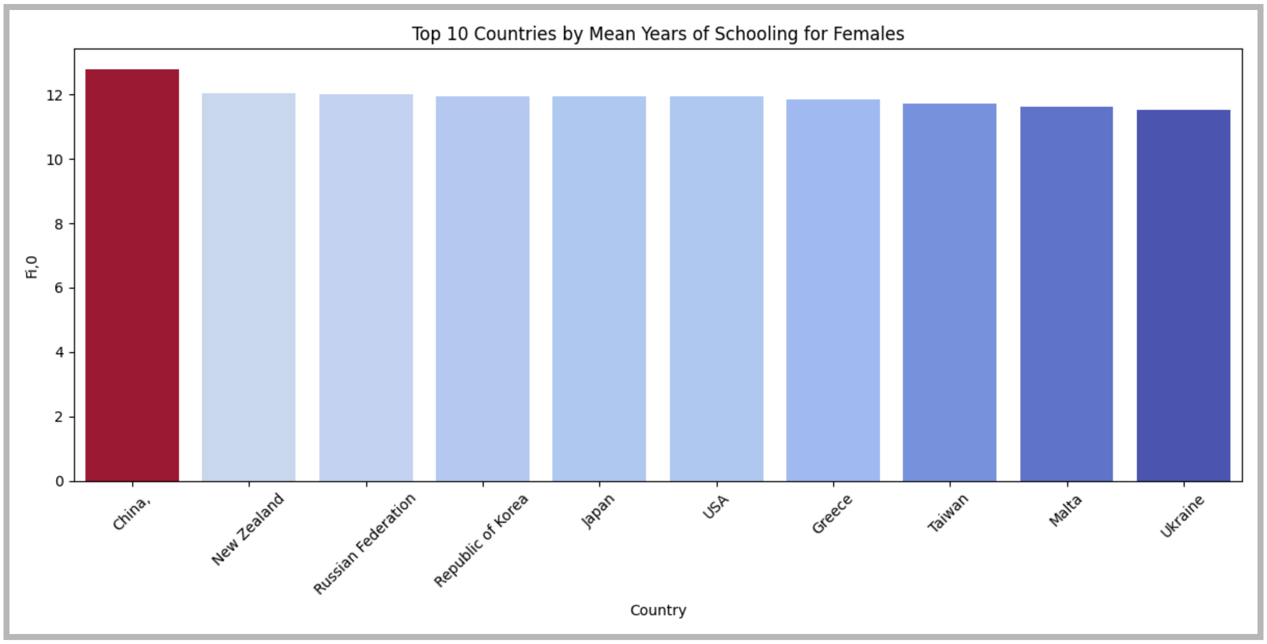
	Fi,0	Mi,0	MFi,0	GBSi,0
Mean	7.407068	7.683253	7.545260	1.151644
Median	8.006500	7.772000	7.752000	0.995808
Std	3.020302	2.448010	2.694922	0.435037
Maximum	12.781000	12.454000	12.612000	4.114488
Minimum	0.953000	1.485000	1.202000	0.725525

The average years of schooling for males has increased to 8.7 years in 2010 from 7.6 years in 1990. We also see similar trends in female schooling years as well. The average years of schooling for females was 7.4 years in 1990 but reached 8.9 years in 2010. Although we see a general trend of increased schooling years across all levels, we also see a slightly higher level of standard deviation in 1990 meaning the inequality has reduced a little over the years till 2010 but not significantly.

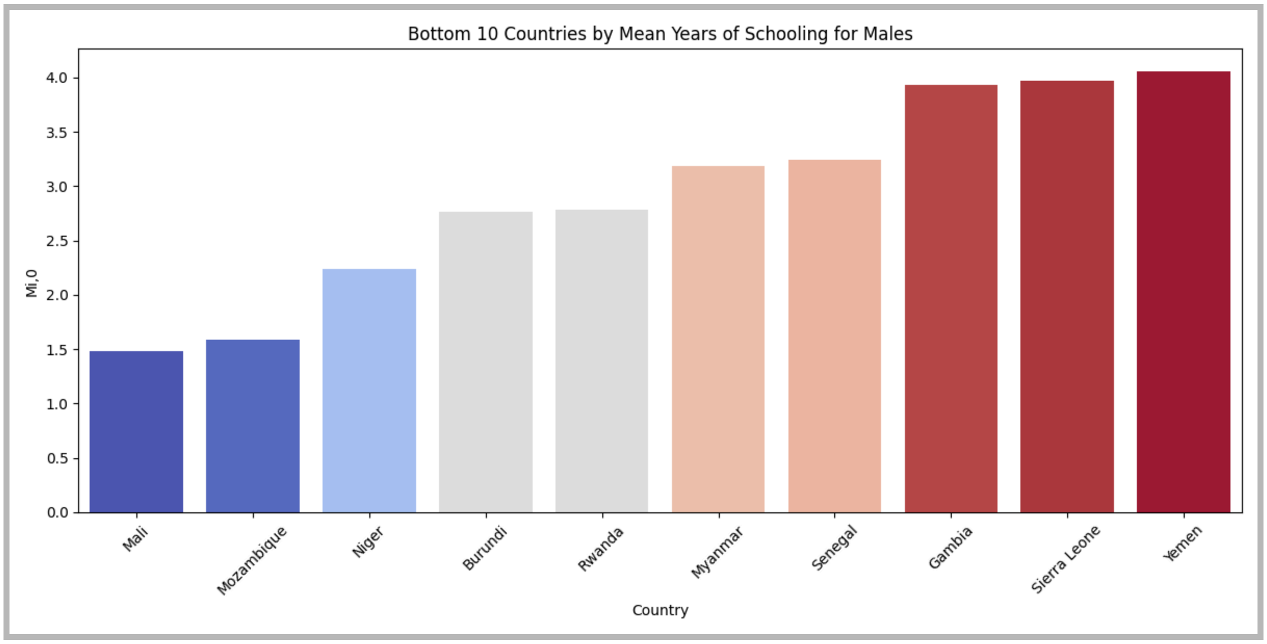
The Maximum values for Males, females and All persons are 12.45,12.78 and 12.61 years of schooling respectively. It is slightly lower compared to the maximum values of average years from 2010. This means the year of schooling has increased over the years. The minimum values of average years of schooling for 1990 is 1.48,0.98 and 1.2 respectively. This data is significantly lower compared to the minimum average years of schooling. Leading to the conclusion that, there is a significant growth over the year for both and female education.

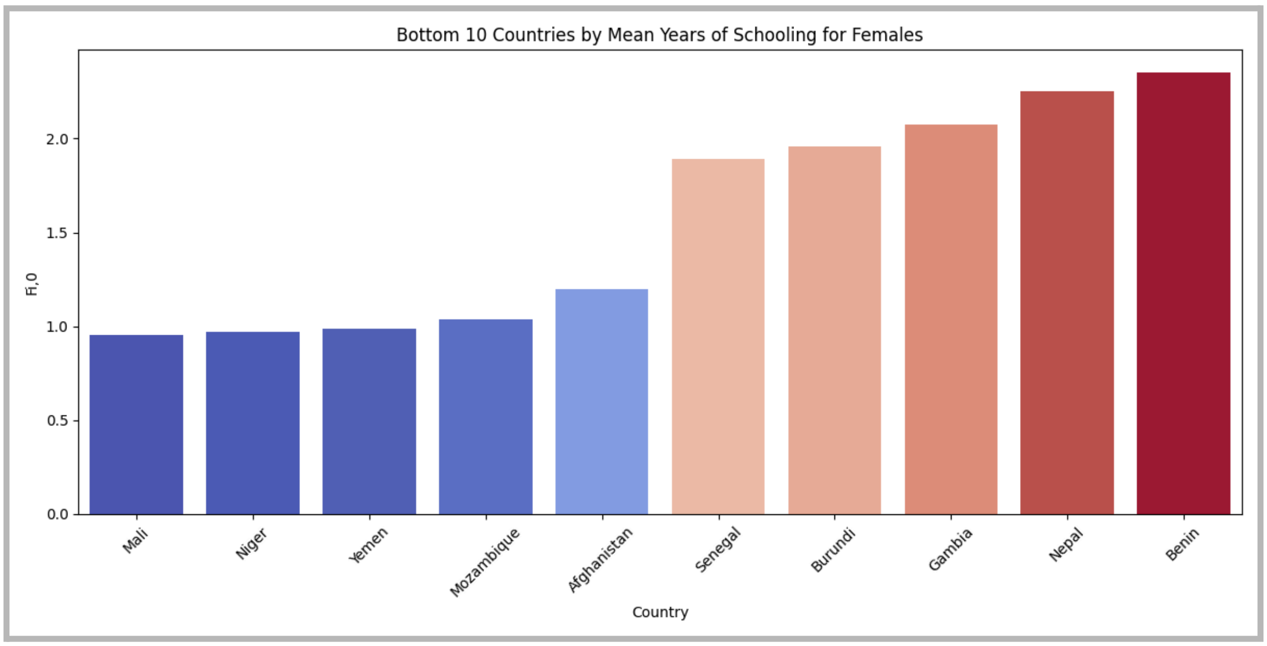
On average, males had slightly higher mean years of schooling compared to females in 1990, suggesting a slight gender bias in favor of males. The gender bias ratio (GBSi,0) being greater than 1 on average suggests a general preference for male education across countries in 1990. However, the median values indicate that half of the countries had almost balanced gender ratios or slightly in favor of females.



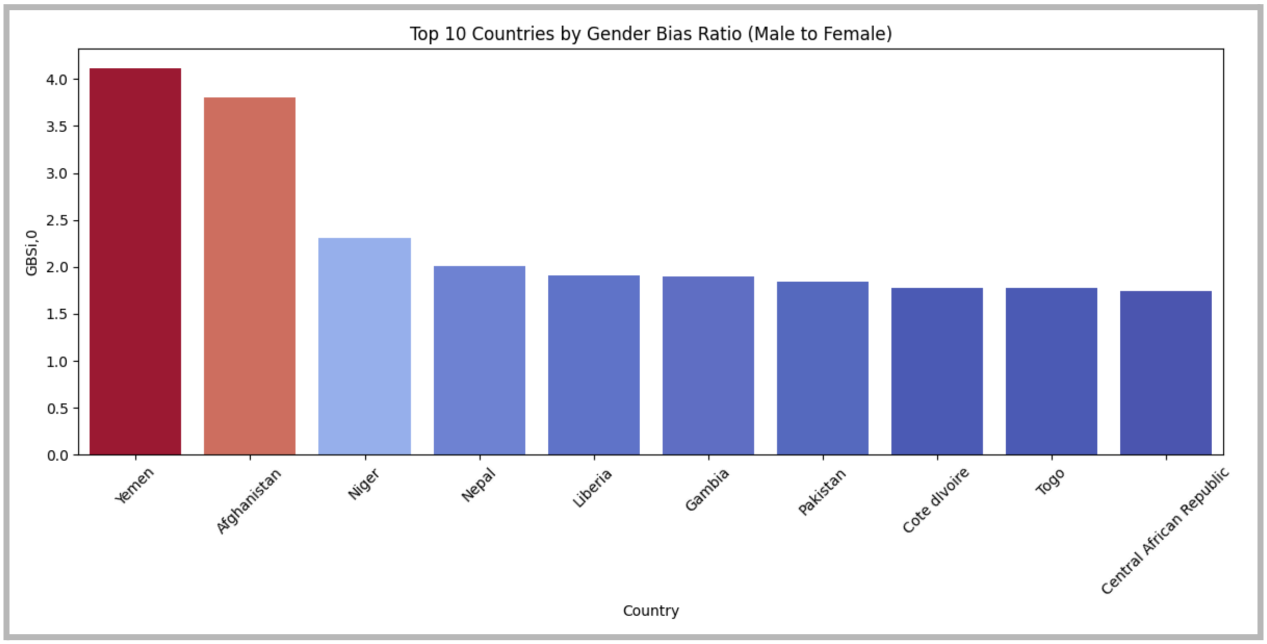


The bar chart represents the countries with the highest average years of schooling for Males and Females in the year 1990. The chart has China, Korea, Japan and Australia showing the regional emphasis of education from the Asia-Pacific region. We can also find countries like Malta, New Zealand, Greece, Russia, USA indicating the strong infrastructure and policy making of the developed nations of Europe and North America during that time. Countries with high average years of schooling often have strong economic policies that support education ultimately leading to economic growth and development.



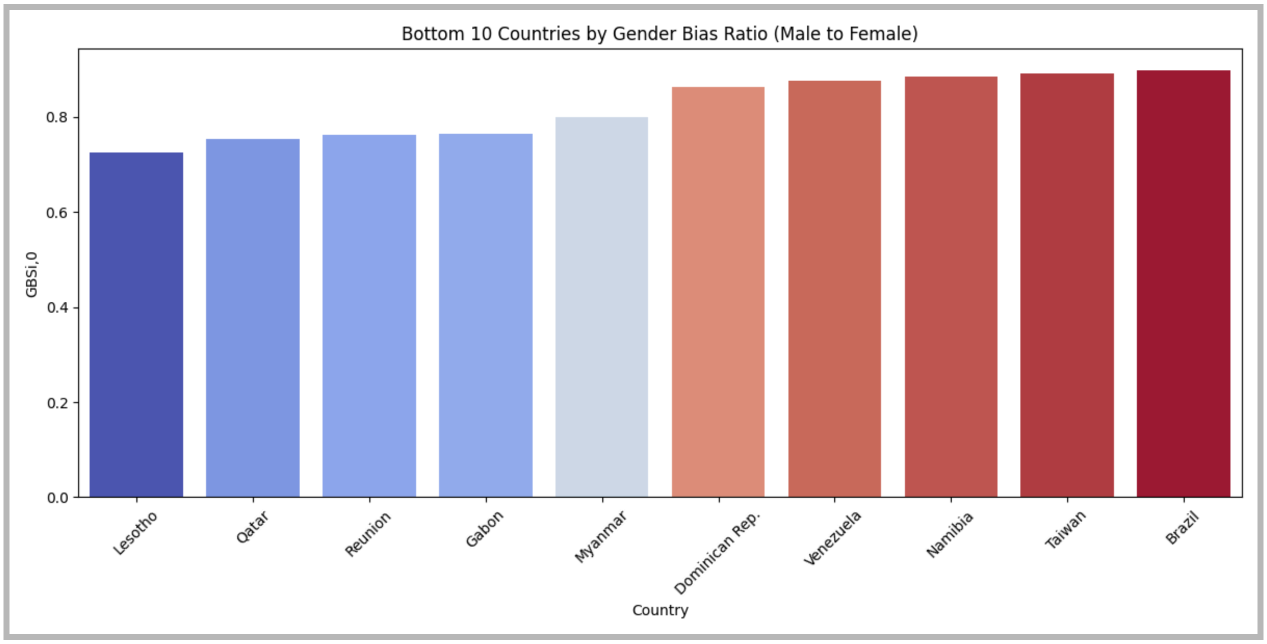


The above 2 bar chart represents the lowest average years of schooling for males globally in 1990. From the chart we can see that it is dominated by countries like Mali, Niger, Yemen, Mozambique, Senegal which are African countries. It was similar for both male and female populations of African countries in 1990. If we compare our results with 1990 and 2010 then we can see similar trends of the African region showing lower mean years of education level. The presence of countries like Myanmar and Yemen indicates that educational challenges were not limited to Africa but were also present in other regions.

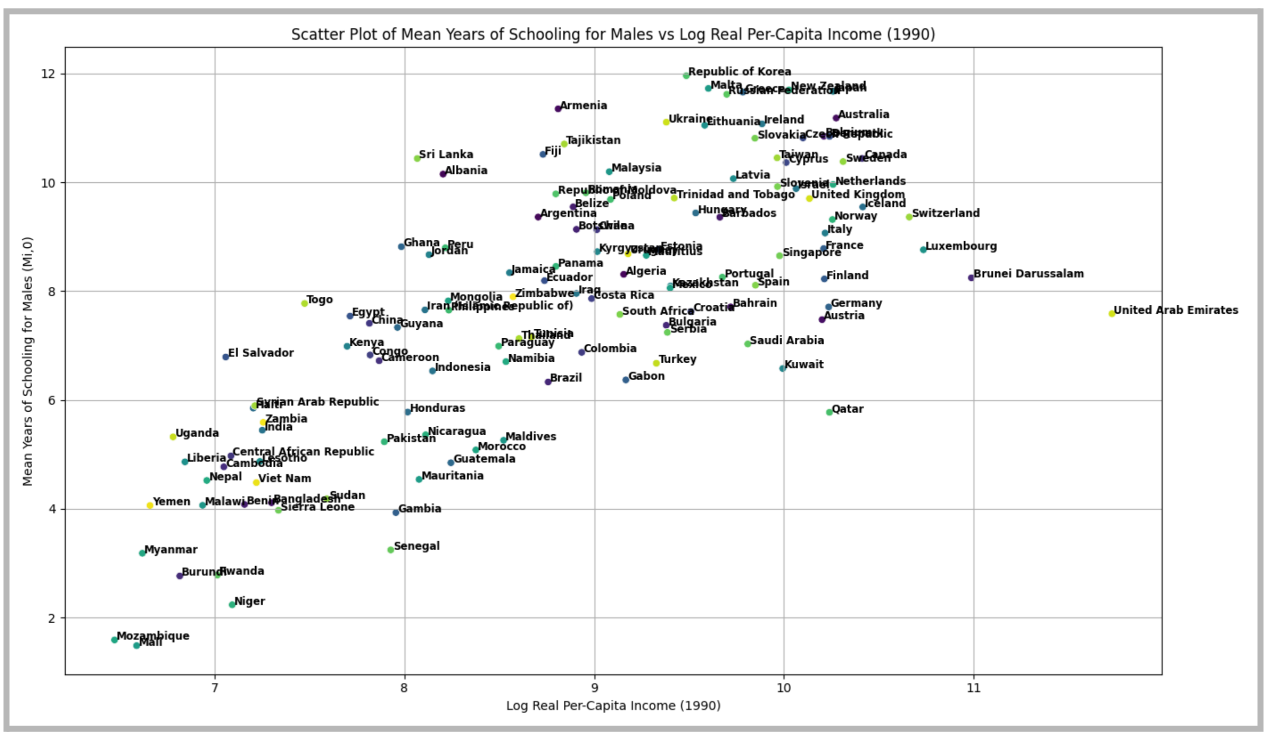


The bar chart represents countries that show significant gender bias in favor of males in 1990. Yemen had the highest gender bias ratio, indicating that males had substantially more years of schooling compared to females. The chart includes countries from various regions, such as the Middle East (Yemen), South Asia (Afghanistan, Nepal, Pakistan), and Africa (Niger, Liberia, Gambia, Cote d'Ivoire, Togo, Central African Republic). This suggests that gender bias in education was not limited to a particle region rather a global issue in 1990.

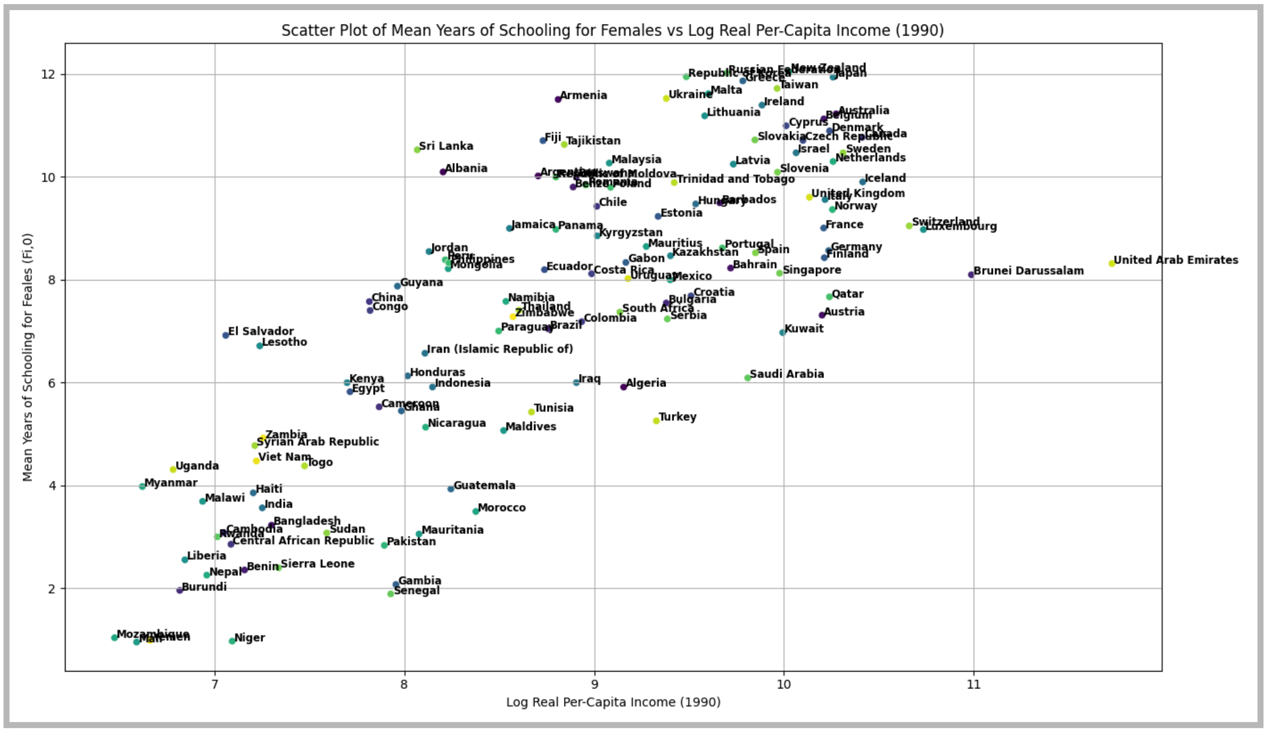
Comparing the results from 1990 to 2010 shows that the gender bias in 1990 was much more prevalent. Historically some countries had social restrictions against female education which might have contributed to such a high number of disparity among males to females. But now the gender bias is declining as we compare to the data from 2010.



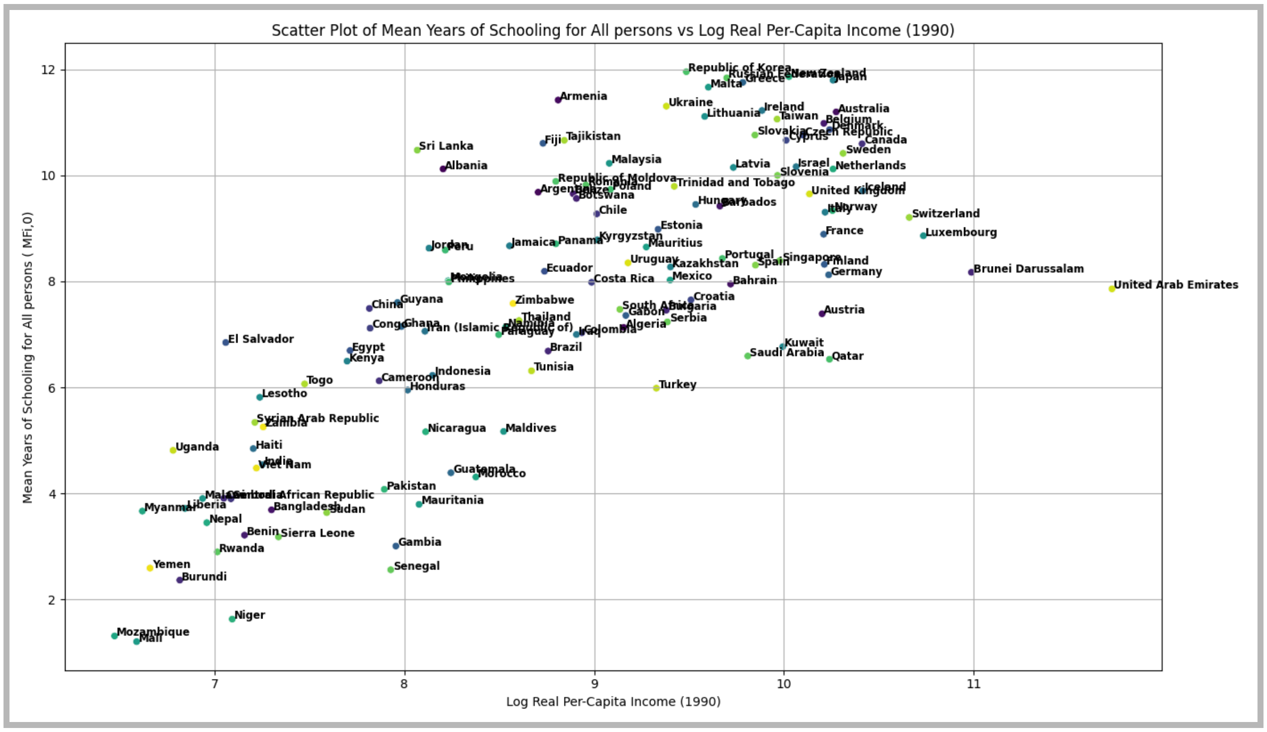
This Bar chart represents a gender bias in favour of females in 1990. Lesotho had the highest bias towards females, on the other hand there countries around the world showing gender bias toward females. The gender bias for females does not belong to any particular region as the countries listed in the chart are from different region.



This scatterplot shows the relationship between the average years of schooling for male and real income in 1990. We can see there is a positive correlation between these two variables.



This scatterplot above shows the relationship between the average years of schooling for female and real income in 1990. We can see there is a positive correlation between these two variables similar to the male population.



The scatterplot above shows the relationship between real per capita income in 1990 and mean years of schooling. Countries like the United Arab Emirates, Kuwait, and Brunei Darussalam have high per-capita income and relatively high mean years of schooling. On the other hand, countries like Mozambique, Mali, and Niger have low per-capita income and low mean years of schooling.

Countries with high per-capita income, typically from regions like North America, Europe, and parts of Asia, show higher mean years of schooling. Countries with low per-capita income, often from regions like Sub-Saharan Africa, show lower mean years of schooling. There is a positive correlation between mean years of schooling and per capita income as we can see from the scatterplot that countries with high levels of per capita income lead to high average years of schooling. Also we can say that higher per capita income can result in more spending in the education sector and resulting in more years in education attainment. And education can lead to more per capita income as well.

(iii) Relationship between income and indicators of schooling using the OLS for All persons in 2010:

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                 LogMFi   R-squared:                       0.518
Model:                            OLS   Adj. R-squared:                  0.514
Method:                 Least Squares   F-statistic:                     136.2
Date:                Thu, 17 Jul 2025   Prob (F-statistic):           7.75e-22
Time:                        00:50:14   Log-Likelihood:                 14.598
No. Observations:                 129   AIC:                            -25.20
Df Residuals:                     127   BIC:                            -19.48
Df Model:                           1
Covariance Type:            nonrobust
===================================================================================
                      coef    std err          t      P>|t|      [0.025      0.975]
-----------------------------------------------------------------------------------
const               0.4128      0.149      2.767      0.007       0.118       0.708
Logrealppi_2010     0.1848      0.016     11.672      0.000       0.153       0.216
==============================================================================
Omnibus:                       32.943   Durbin-Watson:                   2.064
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               55.326
Skew:                          -1.190   Prob(JB):                     9.68e-13
Kurtosis:                       5.152   Cond. No.                         74.2
==============================================================================
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The table above shows the results of the given regression model. The regression model explores the relationship between income and indicators of schooling for all persons. Here the dependent variable is the mean years of schooling in log form from the year 2010 and the independent variable is the per capita income also in natural logarithm form.

The coefficient value of 0.1848 suggests that for a 1% increase in real per-capita income, the mean years of schooling for all persons increases by approximately 0.1848%.The coefficient is highly significant with a p-value of 0.000. So, there is a positive relationship between real per-capita income and mean years of schooling. The model also looks like a good fit for the data as it has a R-squared value of 0.518. The regression analysis suggests a significant positive relationship between real per-capita income and mean years of schooling.

Relationship between income and indicators of schooling using the OLS for Males in 2010:

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  LogMi   R-squared:                       0.495
Model:                            OLS   Adj. R-squared:                  0.491
Method:                 Least Squares   F-statistic:                     124.3
Date:                Thu, 17 Jul 2025   Prob (F-statistic):           1.51e-20
Time:                        01:16:45   Log-Likelihood:                 20.313
No. Observations:                 129   AIC:                            -36.63
Df Residuals:                     127   BIC:                            -30.91
Df Model:                           1
Covariance Type:            nonrobust
===================================================================================
                      coef    std err          t      P>|t|      [0.025      0.975]
-----------------------------------------------------------------------------------
const               0.5572      0.143      3.904      0.000       0.275       0.840
Logrealppi_2010     0.1688      0.015     11.149      0.000       0.139       0.199
==============================================================================
Omnibus:                       35.362   Durbin-Watson:                   2.058
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               64.672
Skew:                          -1.223   Prob(JB):                     9.05e-15
Kurtosis:                       5.460   Cond. No.                         74.2
==============================================================================
```

The table above shows the result of the regression model given for male population. The regression model explores the relationship between income and indicators of schooling for males in 2010. The dependent variable in this regression is (LogMi), which represents the natural logarithm of the mean years of schooling for males in a country in the year 2010 and the independent variable is the per capita income in 2010 also in natural logarithm form.

The coefficient value of 0.1688 suggests that for a 1% increase in real per-capita income, the mean years of schooling for males increases by approximately 0.1688%.The standard errors for the coefficients are relatively small.This regression suggests a significant positive relationship between real per-capita income and mean years of schooling for male as well.

Relationship between income and indicators of schooling using the OLS for Females in 2010:

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  LogFi   R-squared:                       0.523
Model:                            OLS   Adj. R-squared:                  0.519
Method:                 Least Squares   F-statistic:                     139.2
Date:                Thu, 17 Jul 2025   Prob (F-statistic):           3.79e-22
Time:                        01:16:05   Log-Likelihood:                 2.7950
No. Observations:                 129   AIC:                            -1.590
Df Residuals:                     127   BIC:                             4.130
Df Model:                           1
Covariance Type:            nonrobust
===================================================================================
                      coef    std err          t      P>|t|      [0.025      0.975]
-----------------------------------------------------------------------------------
const               0.2292      0.164      1.402      0.163      -0.094       0.553
Logrealppi_2010     0.2047      0.017     11.798      0.000       0.170       0.239
==============================================================================
Omnibus:                       32.005   Durbin-Watson:                   2.083
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               53.980
Skew:                          -1.151   Prob(JB):                     1.90e-12
Kurtosis:                       5.177   Cond. No.                         74.2
==============================================================================
```

The table above shows the results of the regression model given but for the female population. The dependent variable in this regression is (LogFi), which represents the natural logarithm of the mean years of schooling for females in countries in the year 2010.and the independent variable is the per capita income in 2010 also in natural logarithm form.

The coefficient value of 0.2047 suggests that for a 1% increase in real per-capita income, the mean years of schooling for females increases by approximately 0.2047%. The significant F-statistic and coefficients indicates that the model is robust and the findings are statistically significant.The R-squared value of 0.523 indicates that the model explains a big portion of the variance in schooling levels for females.

The regression analysis suggests a significant positive relationship between real per-capita income and mean years of schooling for females similar to what we have found for both males and all persons average years of schooling.

Relationship between income and indicators of gender bias using the OLS in 2010:

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                LogGBSi   R-squared:                       0.184
Model:                            OLS   Adj. R-squared:                  0.178
Method:                 Least Squares   F-statistic:                     28.65
Date:                Thu, 17 Jul 2025   Prob (F-statistic):           3.91e-07
Time:                        11:42:24   Log-Likelihood:                 125.67
No. Observations:                 129   AIC:                            -247.3
Df Residuals:                     127   BIC:                            -241.6
Df Model:                           1
Covariance Type:            nonrobust
===================================================================================
                      coef    std err          t      P>|t|      [0.025      0.975]
-----------------------------------------------------------------------------------
const               0.3280      0.063      5.201      0.000       0.203       0.453
Logrealppi_2010    -0.0358      0.007     -5.353      0.000      -0.049      -0.023
==============================================================================
Omnibus:                       11.021   Durbin-Watson:                   2.128
Prob(Omnibus):                  0.004   Jarque-Bera (JB):               23.509
Skew:                           0.250   Prob(JB):                     7.85e-06
Kurtosis:                       5.031   Cond. No.                         74.2
==============================================================================
```

The table above shows the results of the regression model but given for the female population. The dependent variable in this regression is (LogGBSi), which represents the natural logarithm of the gender bias in schooling.The independent variable (Logrealppi_2010) is the natural logarithm of real per-capita income in 2010. The coefficient value of -0.0358 suggests that for a 1% increase in real per-capita income, the gender bias in schooling decreases by approximately 0.0358%.

This suggests that higher income levels are associated with a reduction in gender bias in schooling, indicating more equal educational attainment between males and females. However, the R-squared value of 0.184 indicates that the model explains a small portion of the variance in gender bias in schooling.

The regression analysis suggests a significant negative relationship between real per-capita income and gender bias in schooling.

(2) Relationship between income and indicators of schooling for All persons in 1990:

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                 LogMFi   R-squared:                       0.561
Model:                            OLS   Adj. R-squared:                  0.558
Method:                 Least Squares   F-statistic:                     162.5
Date:                Thu, 17 Jul 2025   Prob (F-statistic):           1.78e-24
Time:                        11:44:57   Log-Likelihood:                -29.818
No. Observations:                 129   AIC:                             63.64
Df Residuals:                     127   BIC:                             69.36
Df Model:                           1
Covariance Type:            nonrobust
===================================================================================
                      coef    std err          t      P>|t|      [0.025      0.975]
-----------------------------------------------------------------------------------
const              -0.6599      0.206     -3.204      0.002      -1.067      -0.252
LogRealppi_1990     0.2956      0.023     12.747      0.000       0.250       0.342
==============================================================================
Omnibus:                       23.632   Durbin-Watson:                   1.814
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               32.700
Skew:                          -0.966   Prob(JB):                     7.93e-08
Kurtosis:                       4.535   Cond. No.                         68.4
==============================================================================
```

The table above shows the result of the given regression model for all persons in 1990. The dependent variable in this regression is (LogMFi), which represents the natural logarithm of the mean years of schooling for all persons in a country in the year 1990. The independent variable(LogRealppi_1990) is the natural logarithm of real per-capita income in 1990. The coefficient value of 0.2956 suggests that for a 1% increase in real per-capita income, the mean years of schooling for all persons increases by approximately 0.2956%.

The regression analysis suggests a significant positive relationship between real per-capita income and mean years of schooling for all persons. If we compare our results with 2010 then we can see similar trends in both results.

Relationship between income and indicators of schooling for Males in 1990:

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  LogMi   R-squared:                       0.530
Model:                            OLS   Adj. R-squared:                  0.526
Method:                 Least Squares   F-statistic:                     143.1
Date:                Thu, 17 Jul 2025   Prob (F-statistic):           1.51e-22
Time:                        11:48:03   Log-Likelihood:                -15.195
No. Observations:                 129   AIC:                             34.39
Df Residuals:                     127   BIC:                             40.11
Df Model:                           1
Covariance Type:            nonrobust
===================================================================================
                      coef    std err          t      P>|t|      [0.025      0.975]
-----------------------------------------------------------------------------------
const              -0.2006      0.184     -1.091      0.277      -0.564       0.163
LogRealppi_1990     0.2477      0.021     11.960      0.000       0.207       0.289
==============================================================================
Omnibus:                       26.283   Durbin-Watson:                   1.740
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               39.442
Skew:                          -1.013   Prob(JB):                     2.72e-09
Kurtosis:                       4.799   Cond. No.                         68.4
==============================================================================
```

The table above shows the result of the given regression model for male population in 1990. The dependent variable in this regression is (LogMi), which represents the natural logarithm of the mean years of schooling for males in a country in the year 1990. The independent variable is (LogRealppi_1990), which is the natural logarithm of real per-capita income in 1990. The coefficient value of 0.2477 suggests that for a 1% increase in real per-capita income, the mean years of schooling for males increases by approximately 0.2477%. The positive and significant coefficient for (LogRealppi_1990) indicates a positive relationship between real per-capita income and mean years of schooling for males.

Relationship between income and indicators of schooling for Females in 1990:

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  LogFi   R-squared:                       0.557
Model:                            OLS   Adj. R-squared:                  0.554
Method:                 Least Squares   F-statistic:                     159.8
Date:                Thu, 17 Jul 2025   Prob (F-statistic):           3.26e-24
Time:                        11:49:39   Log-Likelihood:                -57.749
No. Observations:                 129   AIC:                             119.5
Df Residuals:                     127   BIC:                             125.2
Df Model:                           1
Covariance Type:            nonrobust
===================================================================================
                      coef    std err          t      P>|t|      [0.025      0.975]
-----------------------------------------------------------------------------------
const              -1.3160      0.256     -5.146      0.000      -1.822      -0.810
LogRealppi_1990     0.3640      0.029     12.640      0.000       0.307       0.421
==============================================================================
Omnibus:                       20.904   Durbin-Watson:                   1.951
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               26.364
Skew:                          -0.928   Prob(JB):                     1.88e-06
Kurtosis:                       4.209   Cond. No.                         68.4
==============================================================================
```

The table above shows the result of the given regression model for female population in 1990. The dependent variable in this regression is (LogFi), which represents the natural logarithm of the mean years of schooling for females in a country in the year 1990. The independent variable is (LogRealppi_1990), which is the natural logarithm of real per-capita income in 1990.

The coefficient value of 0.3640 suggests that for a 1% increase in real per-capita income, the mean years of schooling for females increases by approximately 0.3640%.The positive and significant coefficient for (LogRealppi_1990) indicates a positive relationship between real per-capita income and mean years of schooling for females. It is similar to what we found for both males and all person years of education in 1990.

(2) Relationship between income and indicators of gender bias in 1990:

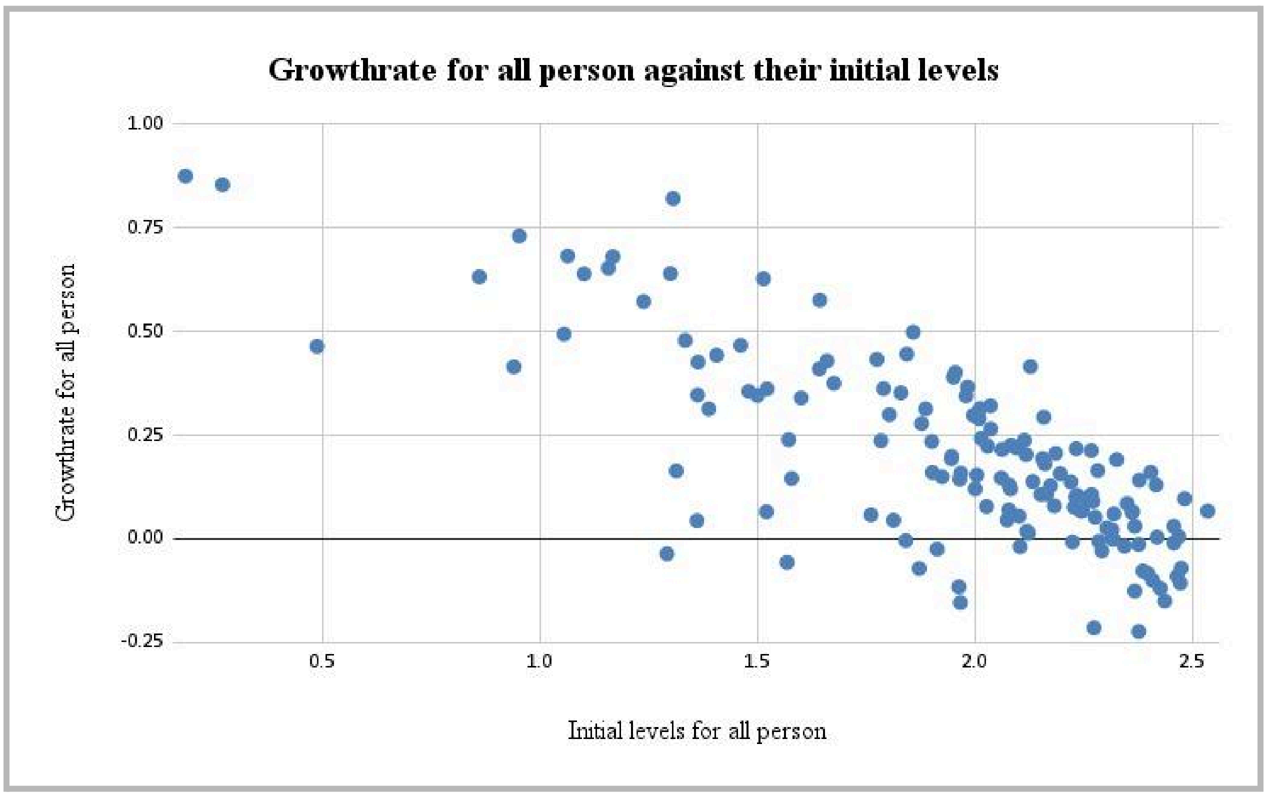
```
                            OLS Regression Results
==============================================================================
Dep. Variable:                LogGBSi   R-squared:                       0.309
Model:                            OLS   Adj. R-squared:                  0.304
Method:                 Least Squares   F-statistic:                     56.85
Date:                Thu, 17 Jul 2025   Prob (F-statistic):           7.84e-12
Time:                        12:04:36   Log-Likelihood:                 22.749
No. Observations:                 129   AIC:                            -41.50
Df Residuals:                     127   BIC:                            -35.78
Df Model:                           1
Covariance Type:            nonrobust
===================================================================================
                      coef    std err          t      P>|t|      [0.025      0.975]
-----------------------------------------------------------------------------------
const               1.1154      0.137      8.141      0.000       0.844       1.387
LogRealppi_1990    -0.1163      0.015     -7.540      0.000      -0.147      -0.086
==============================================================================
Omnibus:                       44.908   Durbin-Watson:                   2.358
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              203.071
Skew:                           1.110   Prob(JB):                     8.01e-45
Kurtosis:                       8.732   Cond. No.                         68.4
==============================================================================
```

The chart above shows the result of a given regression model to explore the relationship between gender bias and real per capita income. The dependent variable in this regression is (LogGBSi), which represents the natural logarithm of the gender bias in schooling in the year 1990. The independent variable (LogRealppi_1990), which is the natural logarithm of real per-capita income in 1990. The coefficient value of -0.1163 suggests that for a 1% increase in real per-capita income, the gender bias in schooling decreases by approximately 0.1163%.

The negative and significant coefficient for LogRealppi_1990 indicates a negative relationship between real per-capita income and gender bias in schooling. This implies that higher income levels are associated with lower gender bias in schooling, implying more equal educational attainment between males and females.

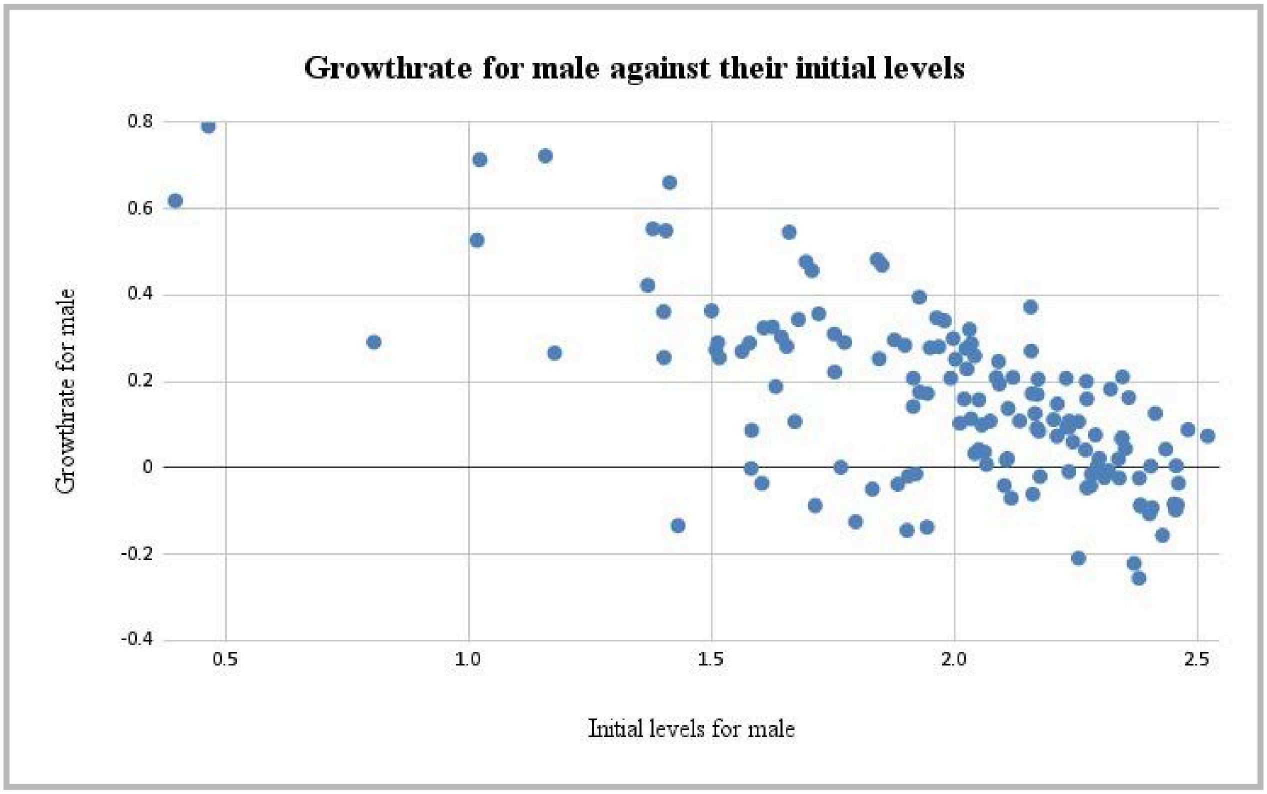
Dynamics of Schooling:

Growth Rate plotting:



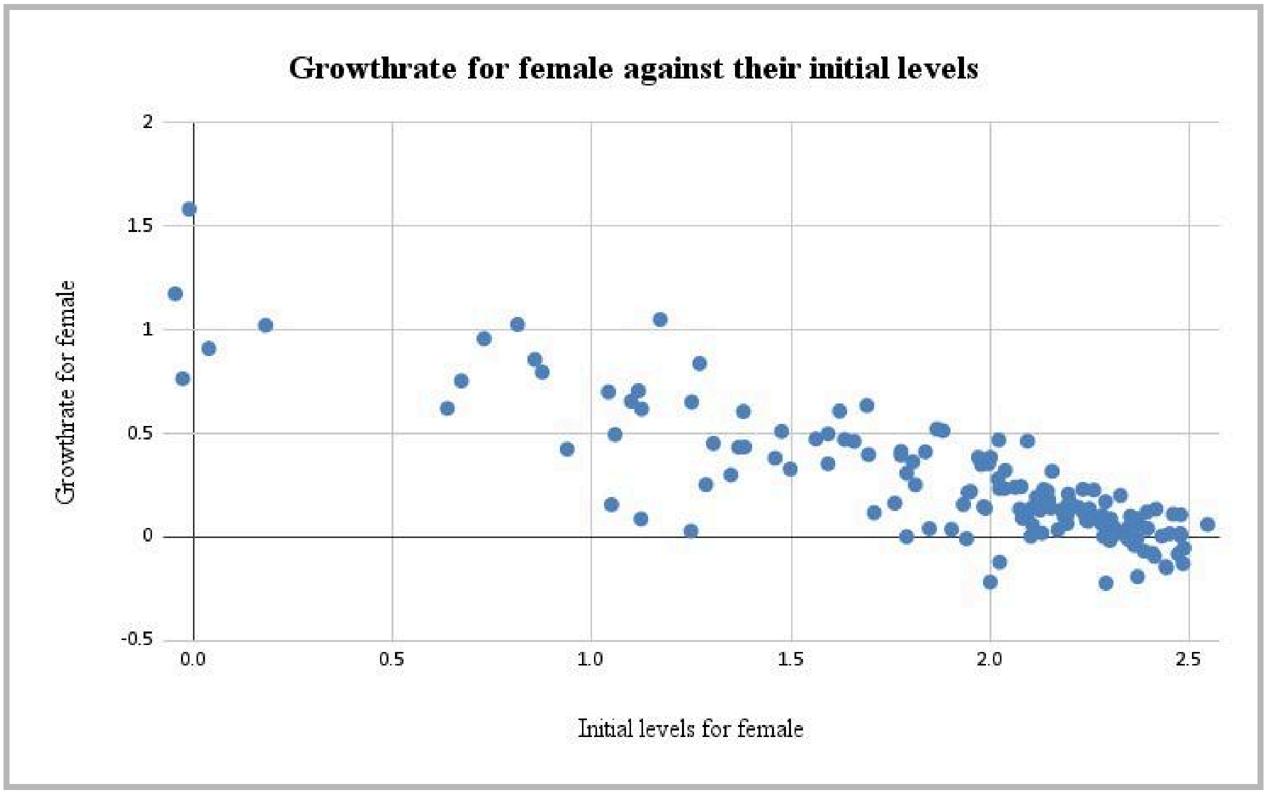
The Scatterplots has an initial level of education in a country in the X-axis and growth rate for all persons in the Y-axis. There is a negative trend in the scatterplot.This means that countries with higher initial levels of schooling tend to have lower growth rates in schooling over the period and countries with lower initial levels of schooling tend to have higher growth rates.The negative relationship seen in the scatter plot could indicate a "catching-up" phenomenon.

Countries that started with lower levels of schooling in 1990 have experienced higher growth rates in schooling by 2010, potentially catching up to countries that had higher initial levels.



This scatterplot is similar to the previous scatterplot showing similar trends. There seems to be a general negative trend in the scatter plot. This suggests that countries with higher initial levels of schooling for males tend to have lower growth rates in schooling over the period. On the other hand, countries with lower initial levels of schooling for males seem to have higher growth rates.

The negative relationship observed in the scatter plot could indicate "catching-up" for males. Countries that started with lower levels of schooling for males in 1990 have seen higher growth rates in schooling by 2010, potentially catching up to countries that had higher initial levels almost similar to the trends seen for all persons.



This scatterplot also seems to have a general negative trend.This trend suggests a trend of convergence in schooling levels for females across countries. Over time, disparities in schooling levels between countries also seem to be narrowing as lower-performing countries improve at a faster rate.

This suggests that countries with higher initial levels of schooling for females seem to have lower growth rates in schooling over the period. Similarly, countries with lower initial levels of schooling for females tend to have higher growth rates.

(iii) (b)Catching up process for all people using regression:

```
                            OLS Regression Results
==============================================================================
Dep. Variable:         Growthrate_Mfi   R-squared:                       0.632
Model:                            OLS   Adj. R-squared:                  0.627
Method:                 Least Squares   F-statistic:                     108.4
Date:                Thu, 17 Jul 2025   Prob (F-statistic):           4.15e-28
Time:                        12:14:53   Log-Likelihood:                 71.172
No. Observations:                 129   AIC:                            -136.3
Df Residuals:                     126   BIC:                            -127.8
Df Model:                           2
Covariance Type:            nonrobust
===================================================================================
                      coef    std err          t      P>|t|      [0.025      0.975]
-----------------------------------------------------------------------------------
const               0.8915      0.098      9.074      0.000       0.697       1.086
LogMFi_1990        -0.4245      0.041    -10.423      0.000      -0.505      -0.344
LogRealppi_1990     0.0148      0.016      0.922      0.358      -0.017       0.047
==============================================================================
Omnibus:                       12.119   Durbin-Watson:                   2.060
Prob(Omnibus):                  0.002   Jarque-Bera (JB):               13.998
Skew:                          -0.601   Prob(JB):                     0.000913
Kurtosis:                       4.078   Cond. No.                         73.7
==============================================================================
```

The table above shows the result of the given regression model. The dependent variable in this regression is (Growthrate_Mfi), which represents the growth rate of the mean years of schooling for all persons from 1990 to 2010.The initial level of schooling in 1990 is (LogMFi_1990). The coefficient value of -0.4245 suggests that for a 1% increase in the initial level of schooling, the growth rate of schooling decreases by approximately 0.4245 percentage points.

The negative and significant coefficient for (LogMFi_1990) shows a negative relationship between the initial level of schooling and the growth rate of schooling. This suggests that countries with higher initial levels of schooling tend to have lower growth rates in schooling over the period, supporting the catching-up hypothesis.

But the coefficient for (LogRealppi_1990) is not statistically significant, indicating that real per-capita income does not have a significant impact on the growth rate of schooling in this model.

The regression result suggests that there is a significant negative relationship between the initial level of schooling and the growth rate of schooling,which indicates the idea of a catching-up effect in educational attainment across countries over the years.

(b)Catching up process for Males

```
                            OLS Regression Results
==============================================================================
Dep. Variable:           Growthrate_Mi   R-squared:                       0.514
Model:                             OLS   Adj. R-squared:                  0.506
Method:                  Least Squares   F-statistic:                     66.57
Date:                 Thu, 17 Jul 2025   Prob (F-statistic):           1.87e-20
Time:                         12:18:19   Log-Likelihood:                 69.646
No. Observations:                  129   AIC:                            -133.3
Df Residuals:                      126   BIC:                            -124.7
Df Model:                            2
Covariance Type:             nonrobust
===================================================================================
                      coef    std err          t      P>|t|      [0.025      0.975]
-----------------------------------------------------------------------------------
const               0.7540      0.096      7.848      0.000       0.564       0.944
LogMi_1990         -0.4165      0.046     -9.025      0.000      -0.508      -0.325
LogRealppi_1990     0.0257      0.016      1.639      0.104      -0.005       0.057
==============================================================================
Omnibus:                        6.766   Durbin-Watson:                   2.093
Prob(Omnibus):                  0.034   Jarque-Bera (JB):                6.318
Skew:                          -0.488   Prob(JB):                       0.0425
Kurtosis:                       3.472   Cond. No.                         70.7
==============================================================================
```

The negative and significant coefficient for (LogMi_1990) indicates a negative relationship between the initial level of schooling for males and the growth rate of schooling for males. This suggests that countries with higher initial levels of schooling for males tend to have lower growth rates in schooling over the period, supporting the catching-up hypothesis for males which is similar to what we have found in the previous regression.

Catching up process for females:

```
                            OLS Regression Results
==============================================================================
Dep. Variable:          Growthrate_Fi   R-squared:                       0.744
Model:                            OLS   Adj. R-squared:                  0.740
Method:                 Least Squares   F-statistic:                     183.2
Date:                Thu, 17 Jul 2025   Prob (F-statistic):           5.16e-38
Time:                        12:20:17   Log-Likelihood:                 57.796
No. Observations:                 129   AIC:                            -109.6
Df Residuals:                     126   BIC:                            -101.0
Df Model:                           2
Covariance Type:            nonrobust
=================================================================================
                      coef    std err          t      P>|t|      [0.025      0.975]
---------------------------------------------------------------------------------
const               1.0241      0.115      8.887      0.000       0.796       1.252
LogFi_1990         -0.4895      0.036    -13.457      0.000      -0.562      -0.418
LogRealppi_1990     0.0175      0.018      0.988      0.325      -0.018       0.053
==============================================================================
Omnibus:                       12.183   Durbin-Watson:                   2.093
Prob(Omnibus):                  0.002   Jarque-Bera (JB):               21.333
Skew:                          -0.407   Prob(JB):                     2.33e-05
Kurtosis:                       4.818   Cond. No.                         78.0
==============================================================================
```

The negative and significant coefficient for (LogFi_1990) indicates a negative relationship between the initial level of schooling for females and the growth rate of schooling for females. This suggests that countries with higher initial levels of schooling for females tend to have lower growth rates in schooling over the period, supporting the catching-up hypothesis for females.

But the coefficient for (LogRealppi_1990) is not statistically significant, indicating that real per-capita income does not have a significant impact on the growth rate of schooling for females in this model.

In conclusion, the regression result suggests a significant negative relationship between the initial level of schooling for females and the growth rate of schooling for females, supporting the idea of a catching-up effect in educational attainment for females across countries.